

SENSITIVITY OF ATRAZINE PARTITION COEFFICIENT ON PESTICIDE ROOT ZONE MODEL PREDICTION OF SOIL'S ATRAZINE RESIDUE DISTRIBUTION WITHIN THE CROP ROOT ZONE

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Previous applications of the Pesticide Root Zone Model (PRZM) comparing predicted distributions of pesticides in the soil profile to the measured values suggest that PRZM prediction capability may be improved by selecting proper representation of the distribution coefficient values that directly influence atrazine-soil interactions. The objective of this study was to observe the variation in the predicted distribution of atrazine in the soil profile as the coefficient of distribution (K_d) was varied in one or more soil horizons using the PRZM-2 model. The goal was to identify if more attention should be devoted to understanding the atrazine-soil sorption issues better, or whether discrepancies in field data and model output are explained by the difficulties in assessing the preferential transport of atrazine. The PRZM-2 predicted atrazine distribution was compared with measured values for no-till (NT) and conventional-till (CT) corn plots. Results indicate that reducing K_d by 70% in NT plots reduced the predicted atrazine recovery in Horizon 1 (0–10 cm) by 31%, 34%, and 87% for simulations performed 2, 4, and 8 weeks after application (WAA). Reducing K_d in only the upper soil horizon improved predictions for the upper horizon but overpredicted atrazine concentrations in the second horizon. Reducing K_d throughout the profile reduced predicted atrazine concentrations in the upper horizon and increased predicted concentrations in lower horizons. Improvement in the correlation between predicted and measured atrazine concentrations was greatest for NT plots; however, discrepancies between predicted and measured data still exist and may be due to improper characterization of the atrazine degradation or dissipation parameters. These results suggest that although preferential flow appears to influence atrazine distribution, it may not be the only process influencing our ability to model atrazine fate and transport. Future research on atrazine degradation and dissipation under field conditions seems to be necessary to simulate atrazine fate and transport under preferential flow conditions. (Soil Science 2007;172:42–54)

Key words: Atrazine, pesticides, modeling, no-till, conventional-till.

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THE Pesticide Root Zone Model (PRZM) (Carsel et al., 1984), PRZM-2 (Mullins et al., 1993), and PRZM-3 (Carousel et al., 1997) models were designed as management tools. Their objective is to assess the effects of various physical, chemical, and biological processes on the fate and transport of pesticides in the soil within and below the crop root zone. Pesticide

Root Zone Model and PRZM-2 are both one-dimensional, dynamic, compartmental models to simulate the fate and transport of agricultural pesticides in the unsaturated zone. Pesticide Root Zone Model 3 has the added capability of simulating nitrogen cycle soil kinetic processes including tracking nitrogen discharged from a septic tank (Mullins et al., 1993; Carouzel et al., 1997). Several researchers have compared PRZM predictions of atrazine distributions in the soil (Zacharias and Heatwole, 1994; Sadeghi et al., 1995), soil pore water (Smith et al., 1991), and surface water runoff (Ma et al., 2000) to data measured during field experiments. Two of these studies also compared predicted and measured Br^- tracer concentrations (Smith et al., 1991; Zacharias and Heatwole, 1994).

The results of the Br^- comparisons suggest that PRZM is able to effectively simulate the movement of water and nonreactive solutes through the crop root zone. The model, however, does not seem to effectively simulate the transport of atrazine into the lower soil horizons (Smith et al., 1991; Zacharias and Heatwole, 1994). Mueller et al. (1992) compared PRZM predicted distributions of alachlor to field-measured values. The model closely predicted the alachlor concentration in the 0- to 8-cm layer between 5 and 40 days after application. Beyond 12 days after application, the model tended to overpredict the alachlor concentration in the lower layer. This was noticeable in the layers of 8 to 15, 15 to 23, 23 to 30, and 30 to 38 cm. However, the model accurately predicted limited alachlor movement beyond 45 cm. These studies suggest that PRZM is able to simulate the transport of some reactive solutes into the lower soil layers and that the inaccuracies in simulations of atrazine transport may be due to improper description of the compound's interactions with the soil matrix.

Sadeghi et al. (1995) compared atrazine distributions measured in 10-cm layers of the 0- to 50-cm soil profile of conventional-till (CT) and no-till (NT) corn plots to PRZM predictions for 3 consecutive growing seasons. The predictions were closest to the measured distributions for the CT plots. For the NT plots, however, PRZM tended to significantly overpredict the atrazine concentration in the 0- to 10-cm layer, whereas it consistently underpredicted the atrazine concentration in the lower layers. The poor correlation between the model simulations and measured atrazine in

the topsoil of the NT plots compared with the similar comparison for the CT plots was attributed to the higher prevalence of macropore flow in the NT plots (Sadeghi et al., 1995). In contrast, Parrish et al. (1992) reported that PRZM accurately predicted Br^- leaching (peak locations) to a depth of 1 m in a field believed to be affected by preferential flow. In this same study, PRZM predictions for metolachlor and aldicarb total carbamate residue in the soil profile were compared with measured values. For metolachlor, the predicted soil recoveries were within a factor of two for at least 90% of the sampling events over a 3-year period. For aldicarb, predicted soil recoveries were within a factor of two for at least 75% of the sampling events over a 4-year period (Parrish et al., 1992). These studies suggest that PRZM is able to predict the distribution of some solutes in the soil profile better than other solutes. It is possible that the values of the parameters used to describe solute-soil relationships need to be site specific. In addition, they may need to be specific for the tillage method, as the coefficient of distribution (K_d) is known to vary as a function of time (hysteresis) and the soil properties of different layers (i.e., fraction of organic carbon).

The current study evaluated the effect of the atrazine-soil partition coefficient (K_d) on PRZM-2 predictions of atrazine distribution in the soil profile of CT and NT corn plots. We concentrated on K_d because previous studies suggest that the atrazine K_d estimated from batch equilibrium experiments depends on both soil type and equilibration time, with published values ranging from 1.04 to 8.1 L/kg (Moreau and Mouvet, 1997; Grundl and Small, 1993; Beck and Jones, 1996; Chen and Wagenet, 1997). It is possible that when initial atrazine transport is sufficiently fast, local equilibrium will not be obtained between the pore water and soil matrix. Initially allowing rapid transport to the lower soil layers, effectively reducing the atrazine concentration in the first horizon (0–10 cm) and increasing the concentration in the four lower horizons (10–20, 20–30, 30–40, and 40–50 cm). Other researchers have reported a difference in K_d measured from batch equilibrium and transport/column style experiments (Bouchard et al., 1988; Kookana et al., 1993; Shimojima and Sharma, 1995). In these studies, more solute was sorbed during batch equilibrium conditions. Kookana et al. (1993) concluded that sorption occurs at a faster rate

under batch equilibrium conditions, presumably due to shaking and the high solution-soil ratio.

Under field infiltration conditions sorption may occur at a slower rate and use of a K_d value measured by batch equilibrium experiments could overestimate the effects of sorption. The following PRZM-2 simulations were accomplished to identify how the correlation between the measured and predicted atrazine distribution in the soil profile was affected by different values of the K_d model parameter. Measured values of atrazine distribution in the soil profile were obtained from a previous study (Sadeghi and Isensee, 1996). Forty PRZM-2 simulations were performed for both the CT and NT fields, modifying on the value of K_d in the upper (0–10 cm) or all five soil horizons. The predicted atrazine distributions in the soil profile were compared with the measured distribution. The goal was to identify if more attention should be devoted to understanding the atrazine-soil sorption issues better, or whether discrepancies in field data and model output are explained by the difficulties in assessing the preferential transport of atrazine.

MATERIALS AND METHODS

Site Description, Field Operation, and Data Collection

The measured soil atrazine residue concentrations and original atrazine-soil K_d values were obtained during a long-term field study at the USDA's Beltsville Agricultural Research Center, Beltsville, Maryland (Sadeghi and Isensee, 1996). The detailed characterization of this site has been previously reported (Isensee et al., 1990; Sadeghi and Isensee, 1992). In brief, the research site is located in the coastal plain region of Maryland and principally consists of Iuka silt loam (coarse-loamy, siliceous, acid Aquic Udi-fluents) and Hatboro silt loam (fine-loamy, mixed, mesic Typic fluvaquents) soils. The field was divided into four plots; two of the plots used NT, and the other two continued to use CT techniques for corn production. Typically 2 to 3 weeks before planting, CT plots were moldboard plowed and disked to a depth of 25 to 30 cm (Sadeghi and Isensee, 1992; Gish et al., 1991). Corn was planted using a two-row, NT planter (Sadeghi and Isensee, 1994). Atrazine (6-chloro-*N*-ethyl-*N'*-(1-methylethyl)-1,3,5-triazine-2,4-diamine) as a wettable powder was sprayed uniformly across all plots, 1 day after

planting. Pesticide was applied (1.34 kg/ha) as active ingredient, using a tractor-mounted sprayer delivering 280 L/ha of water (210 kPa) with TeeJet 730385 flat-fan nozzles (Isensee and Sadeghi, 1995). The actual application rate (C_o) was calculated by measuring the atrazine deposited on 15-cm-diameter Whatman glass fiber filter papers. The filter papers were randomly placed at 15 locations in each plot before herbicide application, removed immediately after pesticide application, and analyzed for atrazine using the same procedure as for soil samples. Results indicate the actual atrazine application rate was 1184 ($\pm$ 121) g/ha (Sadeghi and Isensee, 1996).

Fifteen soil cores (4.1-cm inner diameter by 50-cm outer diameter) were removed from each plot 2, 4, and 8 weeks after herbicide application (WAA). The holes were backfilled with surface soil from the adjacent control field. The core was divided into five 10-cm sections. Each section was placed in a tared glass bottle, weighed, and stored at -5°C for later analysis.

Extraction and Analysis

Details of the extraction and analysis procedure may be found in Isensee and Sadeghi (1994). From each 10-cm section, a 25-g subsample was shake extracted with 100 mL of methanol:water (4:1, vol/vol) for 1 h and filtered, and the filtrate was reduced to 10 mL in a rotary evaporator. Atrazine was removed from the aqueous phase by solid phase adsorption (C18 cartridges), eluted with ethyl acetate, and analyzed with a Hewlett Packard 5890 gas chromatograph equipped with a nitrogen-phosphorus detector (Sadeghi and Isensee, 1996). Recovery of atrazine from spiked soil samples was 85% with a detection limit of 0.01 $\mu\text{g/L}$ (Sadeghi and Isensee, 1996). Atrazine concentrations were not corrected for this recovery.

Statistical and Data Analysis

Statistical methods commonly used for analysis of Gaussian or normally distributed data are suboptimal for estimating sample mean and other statistical parameters (Parken et al., 1988). If the sample values resembled a normal distribution (e.g., bell-shaped pattern), the general statistical procedure was used; otherwise, the method of unbiased minimum variance estimates was applied. This method is based on log-transformed sample values and is recommended when sample size is small (<1000), and sample values are skewed or not normally distributed.

The means were compared with the simulation values predicted by PRZM for the corresponding soil depth at each sampling time during the growing season.

The total amount of atrazine predicted in each soil horizon (C_{ip}) (g/ha) was compared with the measured values (C_{im}) (g/ha) in the corresponding soil horizon. Both measured and predicted atrazine concentrations were converted from units of milligrams per kilogram to milligrams per hectare by

$$\begin{aligned} \text{mg/ha} &= (\text{mg/kg}) * \rho_b (\text{mg/cm}^3) * d(\text{cm}) \\ &\quad * (10^8 \text{cm}^2/\text{ha}) * (\text{kg}/10^6 \text{mg}) \end{aligned}$$

where ρ_b is the bulk density, and d is the depth of the soil horizon. The measured and predicted concentrations were then converted into measured and predicted recoveries (R_{im} and R_{ip} , respectively) by

$$R_{im} = C_{im}/C_o * 100\% \text{ and } R_{ip} = C_{ip}/C_o * 100\%$$

where the first horizon ($i = 1$) is the uppermost (0–10 cm) layer, and the lowest horizon ($i = 5$) is the 40- to 50-cm layer, and C_o is the actual atrazine application rate (1184 g/ha).

Several methods were used to compare predicted recoveries to the field measured values. For each method, a range of predicted recoveries were identified for each soil horizon at each sampling time. Predicted recovery values that fell within the range were identified. The “goodness of fit” for a particular simulation scenario was evaluated by comparing the number of predicted values within the range of each comparison method. The four comparison methods are described in Table 1. Each comparison method

is a range of values based on the statistics of the measured recovery. The range of the R_{10} method is bounded by a low value that is equal to a factor of 10 below ($R_{im}/10$), whereas the high-value boundary is equal to a factor of 10 above ($R_{im}*10$) the average measured recovery for the soil horizon of interest (R_{im}). The range of the R_2 method follows a similar procedure based on a factor of 2. The $C_{1\%}$ and $C_{10\%}$ methods use the confidence limits of the measured data at α levels of 1% and 10%, respectively, reported in Sadeghi and Isensee (1996), expressed as a percentage of applied concentration.

PRZM Description and Model Parametrization

For this study, the PRZM portion of the PRZM-2 model (Carsel et al., 1984) was used to simulate atrazine transport through the crop root zone and distribution in the underlying soil. Pesticide Root Zone Model 2 links together an updated version of the Pesticide Root Zone Model and the Vadose Zone Flow and Transport Model (Mullins et al., 1993). Pesticide Root Zone Model 2 simulates one-dimensional (downward) water and pesticide transport and pesticide transformation through the crop root zone and underlying unsaturated zone to the first water table. The model’s three basic components are soil profile characterization, hydrologic, and chemical fate and transport. Two data files are required for PRZM-2 simulations: a soil and pesticide parameter file and a climatological information file. For these simulations, daily evaporation was calculated by PRZM-2 from daily temperature data. Measured rainfall was included in the climatic

TABLE 1
Description of methods to compare measured with PRZM-2 predicted recoveries

Method		Description	Range
A	R_{10}	$\pm$ a factor of 10 from the measured recovery (R_{im})	$R_{im}/10 < R_{ip} < R_{im}*10$
B	R_2	$\pm$ a factor of 2 from the measured recovery (R_{im})	$R_{im}/2 < R_{ip} < R_{im}*2$
C	$C_{1\%}$	Between the lower (L) and upper (U) confidence limits at α levels of 1%, expressed as a percentage of applied concentration (C_o) [†]	$L_{1\%} < R_{ip} < U_{1\%}$
D	$C_{10\%}$	Between the lower (L) and upper (U) confidence limits at α levels of 10%, expressed as a percentage of applied concentration (C_o) [†]	$L_{10\%} < R_{ip} < U_{10\%}$

R_{ip} = Predicted Recovery in percentage of applied atrazine, single value result of simulation.
 R_{im} = Measured Recovery in percentage of applied atrazine, spatially averaged value from field measurements.
[†]Confidence limits reported with measured data (Sadeghi and Isensee, 1996).

TABLE 2
Measured values of selected soil properties

Soil horizon (cm)	Bulk density (g/cm ³)	Organic carbon (%)	Sand (%)	Clay (%)
NT				
1 (0–10)	1.45	2.5	36.5	17.2
2 (10–20)	1.53	1.3	36.5	18.3
3 (20–30)	1.62	0.70	34.9	22.2
4 (30–40)	1.67	0.66	35.6	25.8
5 (40–50)	1.74	0.65	38.7	25.4
CT				
1 (0–10)	1.35	1.35	35.2	15.7
2 (10–20)	1.43	1.30	34.2	15.8
3 (20–30)	1.56	0.68	31.7	19.8
4 (30–40)	1.63	0.66	30.6	23.9
5 (40–50)	1.68	0.65	36.5	22.4

information file. The soil compartment size for PRZM-2 was 0.67 cm, atrazine leaching was simulated to a depth of 50 cm, and the total soil profile was also simulated to 50 cm (shortest depth to clay layer). The total soil profile was divided into five distinct horizons. Dispersion and biodegradation were not considered to concentrate on the effects of the variation of K_d on the model predictions. The other major values required for PRZM-2 simulations are shown in Tables 2 and 3.

The difference in tillage methods between NT and CT plots were characterized by their

different values of bulk density, percentage of organic carbon, soil texture, pan factor, runoff curve number, and K_d , as shown in Tables 2–4. A separate weather input file described the daily rainfall amount, average temperature, average wind speed, and solar radiation. Weather data were obtained from a weather station located within 500 m of the field site.

Table 4 shows the distribution of K_d values for the different simulations. All other parameters were kept constant in the simulations. The field K_d simulations used K_d values measured from batch equilibrium analyses of soil samples of the upper three soil horizons from CT and NT plots (Sadeghi et al., 1995). Measurements were not available for the two lower horizons; K_d was assumed to be equal to the lowest measured value. In Prediction A, K_d for the first soil horizon was significantly reduced, whereas the other K_d values remained equal to the field measured values. Predictions B and C used significantly reduced K_d in all of the soil horizons. It was expected that lowered K_d values would reduce the amount of atrazine predicted in the soil layer and increase the amount of atrazine available for accumulation in the lower soil layers. Predictions A, B, and C are just three of the approximately 40 different K_d scenarios simulated. The results of these predictions show the amount of variation in K_d that was necessary to significantly modify the

TABLE 3
Selected PRZM input variables

Variable	NT	CT	Comments
Pan factor	0.76	0.72	
Pan evap. factor	1	1	PRZM estimates from weather data
Surface condition	3	2	3, residue 2, cropping
Erosion flag	0	0	Negligible measured soil erosion
Runoff curve number	80	75	
Minimum depth (cm) of evapotranspiration	15	15	
Maximum active root depth (cm)	50	50	Minimum depth to clay layer
Maximum areal crop coverage (%)	85	85	
Plant uptake efficiency flag	1	1	Transpiration* dissolved conc.
Application rate (g/ha)	1184	1184	Average measured application
Depth of incorporation (cm)	0	0	Sprayed on surface
Dispersion coefficient	0	0	Neglected the effects of dispersion
Biodegradation flag	0	0	Neglected the effects of biodegradation
Soil depth (cm)	50	50	Minimum depth to clay layer
Thickness of horizon (cm)	10	10	
Field capacity flag	1	1	Estimated by model
Method of characteristics flag	0	0	MOC was not applied
Irrigation flag	0	0	No irrigation
Surface albedo	0.15	0.15	

TABLE 4
Partition coefficients (K_d) (L/kg) per soil horizon

Soil horizon (cm)	Simulation identification					
	Field K_d [†]		Prediction A [‡]		Prediction B [‡]	Prediction C [‡]
	CT	NT	CT	NT	CT & NT	CT & NT
(0–10)	0.77	0.99	0.30	0.30	0.30	0.30
(10–20)	0.83	0.73	0.83	0.73	0.10	0.05
(20–30)	0.72	0.56	0.72	0.56	0.10	0.05
(30–40)	0.56	0.56	0.56	0.56	0.10	0.05
(40–50)	0.56	0.56	0.56	0.56	0.10	0.05

[†]Field K_d for soil horizons 0 to 10, 10 to 20, and 20 to 30 cm were taken from Sadeghi et al. (1995). Below the plow layer (30 cm), the K_d value was assumed to be 0.56 L/kg based on the value measured in the 20- to 30-cm horizon of the NT plot. Field measured K_d values were estimated by batch equilibrium experiments performed using soil from the research site.

[‡]Predictions A, B, and C represent 3 of the 40 simulations testing various combinations of simulated K_d values.

predicted atrazine distribution in the soil profile for this field site.

RESULTS AND DISCUSSION

Tables 5 and 6 compare the measured distribution of atrazine recovery (R_{im}) in the soil profile with that predicted by the four different simulations using PRZM-2 (R_{ip}). Figures 1–6 show the average measured atrazine recovery (R_{im}) and the predicted recovery for the four simulations for each soil horizon at 2, 4, and 8 WAA for both the NT and CT plots. Figure 7 shows the daily precipitation recorded for the area starting almost 1 month before pesticide application through the

final soil sampling date (8 WAA). Comparison of the CT and NT plots shows that at each sampling time point (2, 4, and 8 WAA), the measured recovery from Horizon 1 (0–10 cm) was higher in the CT plots than the NT plots. In addition, the recovery from the CT plots decreased dramatically toward a minimum value in the lower horizons (Horizons 2 through 5). The measured recovery in the NT plots shows a slightly different leaching pattern. Except for Horizon 1, the measured recoveries in the lower horizons (Horizons 2–5) tend to be higher than for the corresponding CT horizon. In addition, at each sampling point the NT recovery did not decrease as dramatically with depth compared with the CT data.

TABLE 5
Comparison of field measured (R_{im}) and PRZM predicted (R_{ip}) recovery for NT plots

NT recovery					
Horizon	R _{im}	R _{ip}			
		Field K _d	Prediction A	Prediction B	Prediction C
2 WAA					
1	34.5	73.6	50.5	47.8	47.5
2	7.7	1.5	21.7	18.7	16.9
3	1.8	0	0.1	4.2	5.3
4	5.5	0	0	0.6	1.1
5	1.4	0	0	0.1	0.3
4 WAA					
1	7.6	65.6	43.2	40.6	40.4
2	1.9	1.4	19.8	15.7	13.7
3	0.8	0	0.1	3.7	4.7
4	0.4	0	0	0.5	1.0
5	0.3	0	0	0.1	0.2
8 WAA					
1	2.5	34.2	4.4	3.4	3.4
2	2.0	13.3	36.7	13.6	10.2
3	0.8	0.2	1.3	12.4	11.6
4	1.1	0	0	3.5	5.0
5	0.8	0	0	0.6	1.5

TABLE 6
Comparison of field measured (R_{im}) and PRZM predicted (R_{ip}) recovery for CT plots

CT recovery					
Horizon	R _{im}	R _{ip}			
		Field K _d	Prediction A	Prediction B	Prediction C
2 WAA					
1	53.1	77.3	49.9	46.9	46.6
2	4.1	2.4	26.8	21.7	19.3
3	0.9	0	0.2	5.9	7.2
4	0.3	0	0	1.0	1.7
5	0.3	0	0	0.1	0.4
4 WAA					
1	9.8	68.9	42.7	39.8	39.6
2	1.7	2.2	24.4	18.2	15.7
3	0.8	0	0.2	5.3	6.4
4	0.3	0	0	0.9	1.5
5	0.3	0	0	0.1	0.3
8 WAA					
1	5.1	33.1	3.6	2.8	2.7
2	1.9	17.2	39.6	12.3	9.0
3	1.0	0.3	2.5	14.0	12.3
4	0.4	0	0	5.4	6.9
5	0.5	0	0	1.3	2.5

The measured atrazine recovery in the soil profile (R_{im}) (Sadeghi and Isensee, 1996), seemed to be significantly less at 4 WAA than 2 WAA for all but the lower CT soil layers. Essentially, no rainfall occurred between these two sampling dates; therefore, the loss of atrazine was likely due to dissipation processes such as volatilization, irreversible binding, or biological and chemical degradation, although the authors did not elaborate on other possible mechanisms of atrazine fate in field.

For both CT and NT plots, predicted recovery (R_{ip}) for Horizon 1 using field K_d , are significantly higher than measured values. In Horizon 2 (10–20 cm), for both plots, PRZM-2 initially underpredicts the atrazine recovery at 2 WAA, closely predicts recovery at 4 WAA, and significantly overpredicts recovery at 8 WAA. In the lower horizons (3–5), PRZM-2 consistently underpredicts recovery. The differences are most obvious in the NT plots, especially at the earlier sampling times (2 and 4 WAA).

The PRZM-2 prediction scenarios A, B, and C were made by artificially reducing the atrazine-soil K_d value in one or all soil horizons (Table 3) in attempt to account for nonequilibrium and reduced partitioning. Compared with the field measured K_d values in Horizon 1 (0–10 cm), K_d was reduced by 61% and 70% in the CT and NT simulations. When K_d was reduced only in Horizon 1, the reduction in the

predicted recovery for Horizon 1 was 35%, 38%, and 89% for the CT plots 2, 4, and 8 WAA respectively. The reductions were slightly less for the NT plots at 31%, 34%, and 87%. The reduction in the predicted recovery of this horizon was approximately 3 percentage points greater when K_d was reduced throughout the soil profile. These observations indicate that PRZM-2 is sensitive to changes in K_d and illustrate the importance of properly characterizing the parameters associated with atrazine retardation. However, large reductions of the estimated value of K_d from that measured by

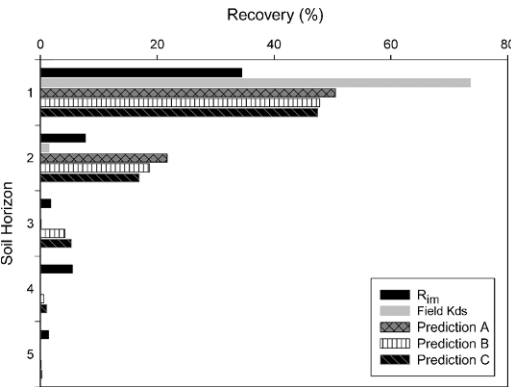


Fig. 1. Measured and PRZM-2 predicted atrazine recovery at 2 WAA in each 10-cm soil horizon of the NT plots.

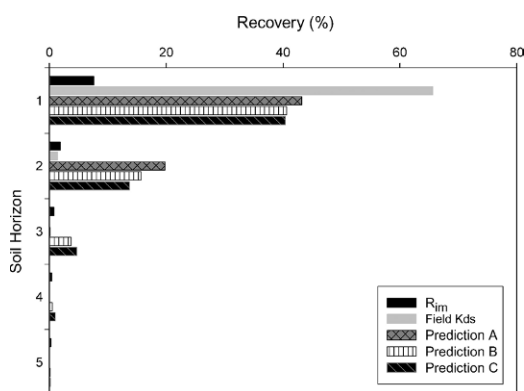


Fig. 2. Measured and PRZM-2 predicted atrazine recovery at 4 WAA in each 10-cm soil horizon of the NT plots.

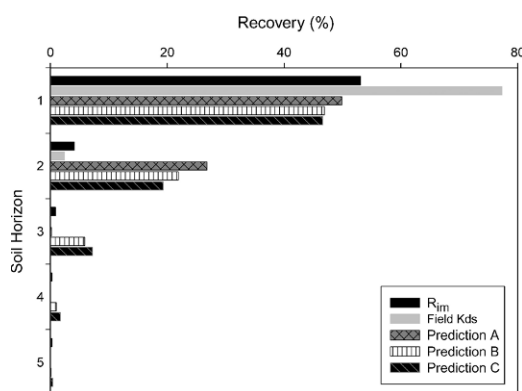


Fig. 4. Measured and PRZM-2 predicted atrazine recovery at 2 WAA in each 10-cm soil horizon of the CT plots.

batch equilibrium experiments were necessary to improve the correlation between the simulated distributions of atrazine recovery and the measured values for both CT and NT plots.

Table 7 summarizes the number of predicted values that fell within the range of each comparison method for each simulation method. The maximum value possible in this table is 15 (3 sampling points [2 WAA, 4 WAA, and 8 WAA] $\times$ 5 horizons) simulation opportunities for each simulation method. The “goodness” of the correlation between predicted and measured recoveries was evaluated by identifying the predicted recoveries that fell within the range of each of the four comparison methods. By comparing the number of predicted recovery values within the range for the CT and NT plots, it seems that the simulations for the NT plots respond more favorably to the

reduction in K_d than the CT plots. These results are most obvious at later sampling points and in lower horizons.

The R_{10} and R_2 comparison methods, a range bounded by a factor of 10 or 2 above and below the average measured value, have been used or described by other researchers comparing measured data with predicted values (Mueller et al., 1992; Parrish et al., 1992; Smith et al., 1991).

Tables 8 and 9 supply the data summarized in Table 7. These tables identify the lower and upper bounds of the range for each comparison method, at each sampling opportunity, for both the NT (Table 8) and CT (Table 9) plots. The PRZM-2 predicted recoveries are also shown for comparison.

For this study, PRZM-2 was not calibrated before use, although many of the input values

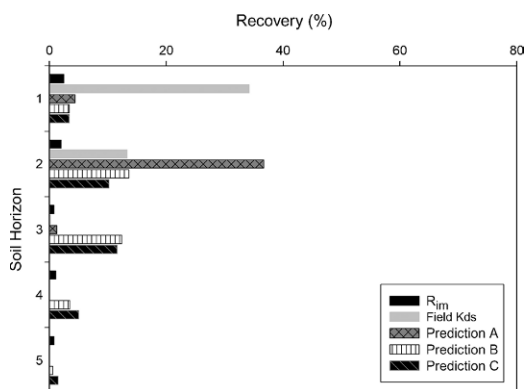


Fig. 3. Measured and PRZM-2 predicted atrazine recovery at 8 WAA in each 10-cm soil horizon of the NT plots.

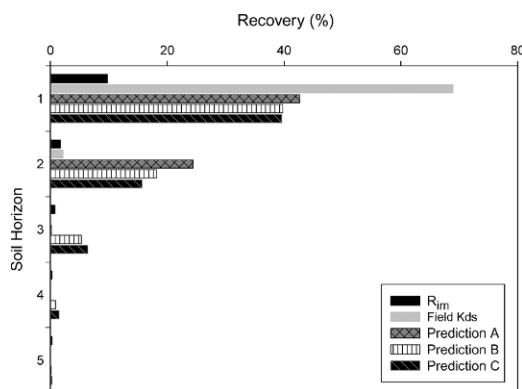


Fig. 5. Measured and PRZM-2 predicted atrazine recovery at 4 WAA in each 10-cm soil horizon of the CT plots.

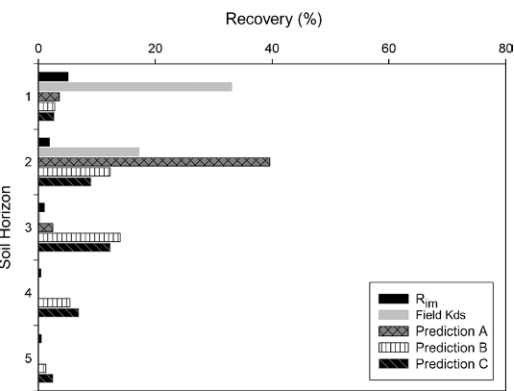


Fig. 6. Measured and PRZM-2 predicted atrazine recovery at 8 WAA in each 10-cm soil horizon of the CT plots.

were based on measurements from field samples. Given the broad range of the R_{10} method, it is not surprising that 12 of the 15 simulation opportunities for the CT plots were within the R_{10} range using the K_d values measured from field soil samples. Only 8 of the 15 NT plot simulation opportunities were within the R_{10} range when the field K_d values were used. This alone suggests that preferential flow processes, not accounted for by PRZM-2, may be responsible for the reduced modeling success of PRZM-2 for the NT compared with CT plots. Using the R_2 comparison method, however, very few of the NT or CT simulation opportunities had predicted atrazine recoveries within the R_2 range, suggesting that for both tillage treatments, characterization of the field condition or atrazine-soil interaction relationship may have been incomplete.

The $C_{1\%}$ and $C_{10\%}$ comparison methods were a range of values bounded by the upper and lower 1% and 10% confidence intervals of the measured values, respectively. Tables 8 and 9 show the intervals for each comparison

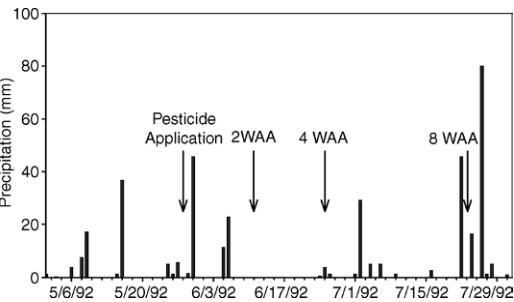


Fig. 7. Daily precipitation.

TABLE 7				
Summary of "goodness of fit"				
Comparison method	Predicted recoveries within range by simulation method			
	Field K_d	Prediction A	Prediction B	Prediction C
NT				
A) $C_{1\%}$	2	5	9	10
B) $C_{10\%}$	1	2	3	2
C) R_{10}	8	8	14	14
D) R_2	1	3	4	4
CT				
A) $C_{1\%}$	4	5	4	5
B) $C_{10\%}$	2	1	1	3
C) R_{10}	12	12	12	13
D) R_2	3	2	2	4

method and the PRZM-2 predicted values for each simulation scenario. Similar comparison methods have been used by Parrish et al. (1992). Using the K_d values measured from field soil samples, PRZM-2 predicted recoveries for the CT plots were slightly more successful than for the NT plots. Reducing the value of K_d throughout the soil profile significantly increased the number of simulations with predicted atrazine recoveries with the $C_{1\%}$ range for NT plots but did not have much of an effect on the predictions for the CT plots. Using the more stringent $C_{10\%}$ comparison method, little change occurred in the number of simulation opportunities that had predicted recoveries within the $C_{10\%}$ range as the value of K_d was reduced in the upper soil horizon (Prediction A) or the entire soil profile (Predictions B and C).

From Tables 5 and 6, it seems that the value of K_d used in Horizon 1 (K_{d1}) significantly influences the distribution of predicted recovery within the soil profile. Reducing K_{d1} reduced the predicted recovery in Horizon 1 to a value close to the measured recovery but significantly overpredicted the recovery in Horizon 2. Reducing the values of K_{d2} through K_{d5} improved the predicted recovery for Horizons 4 and 5. However, Horizons 2 and 3 were typically still significantly overpredicted. K_d values used in Prediction C were approximately 10 to 40% and 10 to 30% of the field measured K_d for the CT and NT plots, respectively. The high predicted recovery that occurred despite exceedingly low K_d values suggests that other important atrazine transport or degradation parameters were not properly considered during this application of PRZM-2.

TABLE 8
Comparison method intervals for NT simulations

Layer	R _{im}	L _{1%}	U _{1%}	L _{10%}	U _{10%}	%		R _{im} *10		R _{im} /2	R _{im} *2	PRZM-2 predicted recovery (%)					
						R _{im} /10	R _{im} *10	R _{im}	Field K _d			Prediction A	Prediction B	Prediction C			
1	34.5	17.3	157.2	23.6	72.0	3.45	345	2 WAA					34.5	73.6	50.5	47.8	47.5
2	7.7	4.1	28.6	5.5	14.5	0.77	77	3.85	15.4	69	7.7	1.5	21.7	18.7	16.9		
3	1.8	1.1	8.1	1.5	4.1	0.18	18	0.9	3.6	3.6	1.8	0.0	0.1	4.2	5.3		
4	5.5	0.8	9.0	1.2	3.9	0.55	55	2.75	11	5.5	5.5	0.0	0.0	0.6	1.1		
5	1.4	0.6	13.3	0.9	4.1	0.14	14	0.7	2.8	1.4	0.0	0.0	0.0	0.1	0.3		
1	7.6	3.5	48.4	4.9	18.6	0.76	76	3.8	15.2	7.6	65.6	43.2	40.6	40.4			
2	1.9	0.8	17.0	1.2	5.6	0.19	19	0.95	3.8	1.9	1.4	19.8	15.7	13.7			
3	0.8	0.4	2.3	0.6	1.3	0.08	8	0.4	1.6	0.8	0.0	0.1	3.7	4.7			
4	0.4	0.3	1.1	0.3	0.7	0.04	4	0.2	0.8	0.4	0.0	0.0	0.5	1.0			
5	0.3	0.2	0.8	0.3	0.5	0.03	3	0.15	0.6	0.3	0.0	0.0	0.1	0.2			
1	2.5	1.5	6.4	1.9	4.0	0.25	25	1.25	5	2.5	34.2	4.4	3.4	3.4			
2	2	1.1	6.5	1.4	3.5	0.2	20	1	4	2	13.3	36.7	13.6	10.2			
3	0.8	0.5	1.9	0.7	1.3	0.08	8	0.4	1.6	0.8	0.2	1.3	12.4	11.6			
4	1.1	0.6	3.5	0.8	1.9	0.11	11	0.55	2.2	1.1	0.0	0.0	3.5	5.0			
5	0.8	0.5	1.9	0.7	1.3	0.08	8	0.4	1.6	0.8	0.0	0.0	0.6	1.5			

Summary of PRZM-2 predicted recoveries within comparison range for NT plots				
Method	Field K _d	Prediction A	Prediction B	Prediction C
C _{1%}	2	5	9	10
C _{10%}	1	2	3	2
R ₁₀	8	8	14	14
R ₂	1	3	4	4

TABLE 9
Comparison method intervals for CT simulations

Layer	%										PRZM-2 predicted recovery (%)			
	R _{im}	L _{1%}	U _{1%}	L _{10%}	U _{10%}	R _{im/10}	R _{im*/10}	R _{im/2}	R _{im*/2}	R _{im}	Field K _d	Prediction A	Prediction B	Prediction C
1	53.1	24.5	319.8	34.9	126.8	5.31	531	26.55	106.2	53.1	77.3	49.9	46.9	46.6
2	4.1	1.7	40.4	2.5	12.5	0.41	41	2.05	8.2	4.1	2.4	26.8	21.9	19.3
3	0.9	0.5	4.5	0.7	2.0	0.09	9	0.45	1.8	0.9	0.0	0.2	5.9	7.2
4	0.3	0.2	1.0	0.3	0.6	0.03	3	0.15	0.6	0.3	0	0	A,C	1.7
5	0.3	0.2	0.7	0.3	0.4	0.03	3	0.15	0.6	0.3	0	0	0.1	0.4
1	9.8	4.2	74.9	6.3	26.3	0.98	98	4.9	19.6	9.8	68.9	42.7	39.8	39.6
2	1.7	0.8	10.1	1.1	4.0	0.17	17	0.85	3.4	1.7	2.2	24.4	18.2	15.7
3	0.8	0.3	3.1	0.5	1.5	0.08	8	0.4	1.6	0.8	0.0	0.2	5.3	6.4
4	0.3	0.1	0.5	0.2	0.3	0.03	3	0.15	0.6	0.3	0.0	0.0	0.9	1.5
5	0.3	0.2	0.7	0.2	0.4	0.03	3	0.15	0.6	0.3	0.0	0.0	0.1	0.3
1	5.1	3.0	14.8	3.8	8.5	0.51	51	2.55	10.2	5.1	33.1	3.6	2.8	2.7
2	1.9	1.3	4.1	1.5	2.9	0.19	19	0.95	3.8	1.9	17.2	39.6	12.3	9.0
3	1	0.6	3.8	0.8	1.9	0.1	10	0.5	2	1	0.3	2.5	14.0	12.3
4	0.4	0.3	1.3	0.3	0.7	0.04	4	0.2	0.8	0.4	0	0	5.4	6.9
5	0.5	0.3	1.2	0.4	0.8	0.05	5	0.25	1	0.5	0.0	0.0	1.3	2.5

Summary of PRZM-2 predicted recoveries within comparison range for CT plots					
Method	Field K _d		Prediction A		Prediction C
	4	2	Prediction B		
C _{1%}			4	5	5
C _{10%}			1	1	3
R ₁₀	12		12	13	13
R ₂	3		2	4	4

CONCLUSIONS

This modeling effort was designed to analyze the effect of reducing the estimated value of the atrazine-soil K_d in the upper 10-cm soil layer (Horizon 1) and entire 50-cm soil profile (Horizons 1 through 5) on the PRZM-2 predicted recovery of atrazine in the soil profile. The results illustrate that the selection of K_d will affect PRZM-2's predicted distribution of atrazine in the soil profile. For the NT plots, reducing the estimated value of K_d by 70% from the values estimated by batch equilibrium experiments with the field soil (field K_d) reduced the predicted recovery of atrazine in Horizon 1 (0–10 cm) by 31%, 34%, and 87% for the 2, 4, and 8 WAA simulations. However, although PRZM-2 is highly sensitive to variations in the value of K_d , modification of K_d only moderately improves the correlation between measured and predicted distributions.

Results of this study suggest that PRZM-2 does not accurately predict the atrazine distribution in NT or CT soil profiles when the effects of sorption are simulated using field K_d values. Reducing the estimated value of K_d tends to slightly improve the correlation between PRZM-2 predictions and measured atrazine distributions, depending on the comparison method used. The improvements illustrate the importance of selecting a K_d value representative of field conditions. However, the estimated values of parameters describing atrazine degradation and dissipation were not well identified for either CT or NT plots. These results suggest that it may be necessary to improve our estimate of these parameters before the effects of preferential flow are evident. The field data indicate preferential flow influences atrazine distribution in the NT compared with the CT soil profile. The simulation results suggest that it may not be the only process influencing our ability to model atrazine fate and transport under field conditions. Future research on atrazine degradation and dissipation under field conditions seems to be necessary to simulate atrazine fate and transport under preferential flow conditions.

REFERENCES

- Beck, A., and K. Jones. 1996. The effects of particle size, organic matter content, crop residues and dissolved organic matter on the sorption kinetics of atrazine and isoproturon by clay soil. *Chemosphere* 32:2345–2358.
- Bouchard, D. C., A. L. Wood, M. L. Campbell, P. Nkedi-Kizza, and P. S. C. Rao. 1988. Sorption nonequilibrium during solute transport. *J. Contam. Hydrol.* 2:209–223.
- Carousel, R. F., J. C. Imhoff, P. R. Hummel, J. M. Cheplick, and A. S. Donigan, Jr. 1997. PRZM-3, A Model for Predicting Pesticide and Nitrogen Fate in the Crop Root and Unsaturated Soil Zones: User's Manual for Release 3.0. Prepared for: U.S. EPA National Exposure Research Laboratory, Athens, GA.
- Carsel, R., C. Smith, L. Mulkey, and J. Jowise. 1984. User's Manual for the Pesticide Root Zone Model (PRZM): Release 1. EPA-600/3-84-109; Athens, GA, U.S. Environmental Protection Agency.
- Chen, W., and R. Wagenet. 1997. Description of atrazine transport in soil with heterogeneous nonequilibrium sorption. *Soil Sci. Soc. Am. J.* 61:360–371.
- Gish, T. J., A. R. Isensee, R. G. Nash, and C. S. Helling. 1991. Impact of pesticides on shallow groundwater quality. *Trans. Am. Soc. Agric. Eng.* 34(4):1745–1753.
- Grundl, T., and G. Small. 1993. Mineral contributions to atrazine and alachlor sorption in soil mixtures of variable organic carbon and clay content. *J. Contam. Hydrol.* 14:117–128.
- Isensee, A., R. Nash, and C. Helling. 1990. Effect of conventional vs. no-tillage on pesticide leaching to shallow groundwater. *J. Environ. Qual.* 19:434–440.
- Isensee, A., and A. Sadeghi. 1994. Effects of tillage and rainfall on atrazine residue levels in soil. *Weed Sci.* 42:462–467.
- Isensee, A., and A. Sadeghi. 1995. Long-term effect of tillage and rainfall on herbicide leaching to shallow groundwater. *Chemosphere* 30:671–685.
- Kookana, R. S., R. D. Schuller, and L. A. G. Aylmore. 1993. Simulation of simazine transport through soil columns using time-dependent sorption data measured under flow conditions. *J. Contam. Hydrol.* 14:93–115.
- Ma, Q., J. E. Hook, R. D. Wauchope, and C. C. Dowler, et al. 2000. GLEAMS, Opus, PRAM2β, and PRZM3 simulation compared with measured atrazine runoff. *Soil Sci. Soc. Am. J.* 64:2070–2079.
- Moreau, C., and E. Mouvet. 1997. Sorption and desorption of atrazine, deethylatrazine, and hydroxyatrazine by soil and aquifer solids. *J. Environ. Qual.* 26:416–424.
- Mueller, T., R. Jones, P. Bush, and P. Banks. 1992. Comparison of PRZM and GLEAMS computer model predictions with field data for alachlor, metribuzin and norflurazon leaching. *Environ. Toxicol. Chem.* 11:427–436.
- Mullins, J., R. Carsel, J. Scarbough, and A. Ivery. 1993. PRZM-2, A Model for Predicting Pesticide Fate in the Crop Root and Unsaturated Soil Zones: Users Manual for Release 2.0.; EPA/ 600/R-93/046; Athens, GA, U.S. Environmental Protection Agency.
- Parken, T., J. Meisinger, S. Chester, J. Starr, and J. Robinson. 1988. evaluation of statistical estimation methods for lognormally distributed variables. *Soil Sci. Soc. Am. J.* 52:323–329.

- Parrish, R. S., C. N. Smith, and F. K. Fong. 1992. Tests of the pesticide root zone model and the aggregate model for transport of aldicarb, metolachlor, and bromide. *J. Environ. Qual.* 21:685–697.
- Sadeghi, A., and A. Isensee. 1992. Effect of tillage systems and rainfall patterns on atrazine distribution. *Soil. J. Environ. Qual.* 21:464–469.
- Sadeghi, A., and A. Isensee. 1994. Spatial distribution of atrazine residues in soil and shallow groundwater: effect of tillage and rainfall timing. *Agric. Ecosyst. Environ.* 48:67–76.
- Sadeghi, A., A. Isensee, and A. Shirmohammadi. 1995. Atrazine movement in soil: comparison of field observations and PRZM simulations. *J. Soil Contam.* 4:151–161.
- Sadeghi, A., and A. Isensee. 1996. Impact of reversing tillage practices on movement and dissipation of atrazine in soil. *Soil Sci.* 161:390–397.
- Shimajima, E., and M. L. Sharma. 1995. The influence of pore water velocity on transport of sorptive and non-sorptive chemicals through an unsaturated sand. *J. Hydrol.* 164:239–261.
- Smith, M. C., A. B. Bottcher, K. L. Campbell, and D. L. Thomas. 1991. Field testing and comparison of the PRZM and GLEAMS models. *Trans. ASAE* 34:838–847.
- Zacharias, S., and C. Heatwole. 1994. Evaluation of GLEAMS and PRZM for Predicting Pesticide Leaching Under Field Conditions. *Trans. Am. Soc. Agric. Eng.* 37:439–451.